

ROFIQUES SALEHIN

Chandler, AZ
Willing to Relocate

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Phone: 681-285-7209
US Permanent Resident

SUMMARY

- Computational modeling, experimental validation and data assimilation expert with 5+ years of experience in Mechanical/Materials engineering demonstrated by 6 publications in peer-reviewed journals including 3 first-author papers.
- Hands-on experience of working more than 3+ years in the Semiconductor Industry.
- Project management and time management skills leading to successful completion of several complex projects using atomic scale simulations and statistical analysis.
- Strong ability to collaborate and work in a multi-disciplinary team environment.
- Legally authorized to work in the United States.

EDUCATION

M.Eng. in Mechanical Engineering, *Colorado State University, Fort Collins, Colorado, USA.*

Track- Computational Material Science, Expected Completion: May 2022, CGPA: 3.48/4.00

M.S. in Mechanical Engineering, *West Virginia University, Morgantown, West Virginia, USA.*

Track – Colloidal Fluids and Functional Particles, Graduation: July 2018, CGPA: 3.87/4.00

B.S. in Aeronautical Engineering (Major- Aerospace), *Military Institute of Science & Technology, Bangladesh University of Professionals, Dhaka, Bangladesh.*

Track – Fluid Mechanics, Graduation: March 2014, CGPA: 3.66/4.00

WORK EXPERIENCE

Facilities Engineer, Fab Facilities and Operations, Central Engineering, Intel Corporation

(April 2023 – Present)

- Owning xPCW and HVAC Mechanical Facility Systems to facilitate the continuous operation of Intel Fabs.
- Providing Facilities engineering leadership to the site organizations in support of Safety, Reliability, Quality, Environmental and Cost objectives.
- Owning safety within area of influence and ensuring systems operate reliably to avoid manufacturing/facility (or operations) impacts by meeting uptime goals.
- Overlooking projects in close collaboration with Facilities Maintenance Planner/Scheduler (MPS) to meet design criteria, scope, budget, and schedule.
- Oversight of planning, design, reconfiguration, construction, maintenance, and modifications to equipment, machinery, facilities systems, buildings, and structures as required to support business requirements.
- Delivering operational efficiency, maintenance, testing and commissioning solutions.
- Effective collaboration with construction and operations teams to coordinate work in the field.
- Identify and implement opportunities to achieve lowest cost of ownership to key stakeholders while maintaining reliability by performing Failure Mode and Effects Analysis (FMEA).
- Applying engineering skills and tools to deliver results through systematic problem solving, statistical validation of system performance, collaboration with the Site Facility management team/suppliers/key stakeholders and through in-depth understanding of the operating systems to determine and ensure compliance to existing standards, specifications, building codes, and processes.
- Supporting the CS engineering teams, mechanical systems reliability efforts, business continuity program and other efficiency improvement projects.

Substrate Quality & Reliability Engineer, Global Supply Chain, Intel Corporation

(June 2022 – March 2023)

- Developed, applied and maintained quality and reliability standards for processing materials into partially finished or finished products.
- Evaluated the materials, process and techniques used in production to meet the requirement of products and production equipment.
- Specified inspection and testing mechanism, conducted quality assessment (up to and including audits) and performed statistical analysis to determine the responsibility for products and systems that do not meet required specifications.
- Delivered standardization in product qualification, manufacturing quality systems and methods.

- Provided feedback to BU, TD and/or partner engineering groups on design or method and statistical process control procedures including manufacturing systems in HVM.
- Determined reliability requirements and automated screening methodologies of components and systems to achieve company, customer, and any governmental agency reliability objectives.
- Developed new acceleration techniques, analytical tools and quality screens to assure the early identification of potential problems with new products, packaging, processes, and product quality/reliability.
- Responded to customer requests or events as they occur.
- Made recommendations for changes in design or formulation to improve system and or process product reliability.
- Worked as the POC for GSC Substrate NPI Quality and Reliability team on 1274 Co-EMIB products (PVC, RLT, FLS) as well as 1276 EMIB products (GNR, SRF, TFN, WLW).

Graduate Research Assistant, ACME Lab, Colorado State University

(Fall 2018 – Summer 2021)

- Worked on a US Department of Energy (DOE) project for tailoring Carbide Dispersed Steels (CDS) to increase strength and hydrogen tolerance.
- Conducted DFT simulations in High-performance computing (HPC) clusters to determine the ground state energy of Ultra High Temperature Ceramics (UHTCs) containing point defects.
- Calculated Hydrogen trapping in Iron-Carbon alloys to resist Hydrogen Embrittlement (HE) as well as storing hydrogen.
- Worked closely with experimental team to validate the simulation results.
- Performed statistical study for diffusion using metropolis and kinetic Monte-Carlo analysis in group IVB and VB transition metal carbides (TMCs).
- Accomplished a large amount of data analysis using Python, C, C++ and MATLAB coding along with several visualization tools such as VMD, Ovito, Inkscape etc.

Graduate Research Assistant, Statistical Mechanics Lab, West Virginia University

(Summer 2017 – Summer 2018)

- Worked on a European Union Horizon 2020 research and innovation program to implement smart fluids using mechanics in mechanical control of fluid viscosity using variable shaped functional particles.
- Performed MD simulations for colloidal shear thickening fluids.
- Used the concept of Jamming to functional star shaped particles.
- Conducted 3D printing and LASER cutting of functional particles.
- Performed necessary statistical data analysis as well as visualization and movie making.

Trainee Engineer, Engineering Planning Department, BIMAN Bangladesh Airlines

(Nov 2012 – Feb 2013)

- Trained in working several workshops (Base and Engine, Aerospace overhaul, NDT, Sheet metal) along with Maintenance sector (Central engineering, Engineering planning, Quality assurance, Ground support equipment, hanger maintenance).
- Instructed on the reliability of aircraft components and systems to ensure the effectiveness of aircraft maintenance tasks and thus maintain the airworthiness.

TECHNICAL SKILLS

- **Programming Language:** Python, MATLAB, Simulink, C, C++, Visual Basic, LaTeX.
- **HPC Platforms:** MPI, OpenMP, Slurm, Torque.
- **Design and Drawing:** SolidWorks, AutoCAD, Revit, Javafoil.
- **Simulation Software:** Ansys, Abaqus, LAMMPS, VASP, ArcGIS.
- **Experimental Tools:** TEM, SEM, LASER cutting, 3D printing, Atom Probe Tomography.
- **Visualization Tools:** VMD, Ovito, Vesta, p4vasp, Power BI, Tableau, Seeq, Scada.
- **Statistical Analysis Tool:** JMP, SAS, SQL, R.

CERTIFICATIONS

- Intel 6D Certification
- EPA 608 Universal Certification
- Intel L1 Cert: Primary/Secondary Heating Water Systems, Primary/Secondary/Tertiary Chilled Water Systems, Condenser/Cooling Tower Water Systems, Process Cooling Water Systems, Critical Process Cooling Water Systems, Primary/Secondary Heat Recovery Water Systems, Heating, Ventilation and Air Conditioning (HVAC), Exhaust Systems.